Moderately Exponential Time Approximation Algorithms for the Maximum Bounded-degree-d Set Problem*

Maw-Shang Chang and Ling-Ju Hung†
Department of Computer Science and Information Engineering
HungKuang University, Taichung 43302, Taiwan
{mschang,ljhung}@sunrise.hk.edu.tw

Abstract

Given a graph $G = (V, E)$, a vertex set $S \subseteq V$ is called an s-plex if every vertex $v \in S$ is of degree at least $|S| - s$ in $G[S]$. The Maximum s-plex (Max s-plex) *problem is to find an s-plex of maximum cardinality in an input graph. It has applications on finding cohesive subgroups in social networks. A* bounded-degree-d set *S in a graph $G = (V, E)$ is a vertex subset of G such that the maximum degree in $G[S]$ is at most d. The* Maximum Bounded-Degree-d Set *(*Max d-bds*) problem is to find a bounded-degree-d set S of maximum cardinality in G. Both* Max s-plex *and* Max d-bds *are NP-hard problems. A vertex subset S is a bounded-degree-d set in G if and only if S is a $(d+1)$-plex in $\bar{G}$. In this paper, we show that if $P \neq NP$, for all $\epsilon > 0$,* Max d-bds *and* Max s-plex *cannot be approximated with a ratio greater than $n^{\epsilon-1}$ in polynomial time for bounded $d \geq 1$ and $s \geq 2$. Moreover, we design moderately exponential time $1/p$-approximation algorithms to solve the* Max d-bds *problem where p is an integer satisfying $p \geq d + 1$. For $d \geq 2$, our $1/p$-approximation algorithms run in time faster than $O^*(2^{n/p})$ where n is the number of vertices in the input graph.*

1 Introduction

A *bounded-degree-d set S* in an undirected graph $G = (V, E)$ is a vertex subset such that the maximum degree of $G[S]$ is at most d. The Maximum Bounded-Degree-d Set problem is to find a bounded-degree-d set S of maximum size in the input graph $G = (V, E)$. An s-plex S in a graph $G = (V, E)$ is a vertex subset such that for each $v \in S$, $deg_{G[S]}(v) \geq |S| - s$. Note that a vertex subset S is a bounded-degree-d set in G if and only if it is a $(d + 1)$-plex in $\bar{G}$. In the following, we list the definitions of the Maximum-Bounded-Degree-d Set problem and the Maximum s-Plex problem.

Maximum Bounded-Degree-d Set
(Max d-bds)
Input: A graph $G = (V, E)$.
Output: A vertex set $S \subseteq V$ of maximum cardinality such that S is bounded-degree-d set.

Maximum s-Plex (Max s-plex)
Input: A graph $G = (V, E)$.
Output: A vertex set $S \subseteq V$ of maximum cardinality such that S is an s-plex.

The Maximum Bounded-Degree-d Set (Max d-bds) problem is NP-complete because the Maximum $(d+1)$-Plex (Max $(d+1)$-plex) problem is NP-complete [3]. The Minimum Bounded-Degree-d Deletion Set (Min d-bdd) problem is the dual problem of the Max d-bds problem. A series of fixed-parameter algorithms were developed for solving the Min d-bdd problem by taking the size of the solution as the input parameter [21, 16, 20, 12]. Betzler *et al.* [5] showed that when parameterized by treewidth the Min d-bdd problem is W[1]-hard. In the same paper, they showed that the problem is fixed-parameter tractable for the following parameters: (i) the combined parameter treewidth *and* the number of vertices to delete; (ii) the feedback edge set number. Chang and Hung [8] gave moderately exponential time approximation algorithms for Max 1-bds and its related problems. An exact algorithm for Max 1-bds was given in [9] running in time $O^*(1.4658^n)$.

The model s-plex is defined as a degree based relaxed model for finding cohesive subgroups in so-

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cial networks [22]. The MAX s-PLEX problem can be formulated in 0/1 integer programming [2, 3]. It was shown W[1]-hard with respect to the parameter k being the size of s-plexes [16]. Some branch-and-bound algorithms were given for solving the MAX s-PLEX problem based on different upper bounds and lower bounds found by heuristics algorithms [23, 19]. Some graph editing problems were studied on finding a disjoint union of k-plexes [14, 6]. Balasundaram *et al.* [4] referred MAX d-BDS as the problem of finding maximum-cardinality co-$(d+1)$-plexes. Wu and Pei [24] gave an algorithm to enumerate all maximal s-plexes.

In this paper, we show that if $P \neq NP$, for all $\epsilon > 0$, MAX d-BDS and MAX s-PLEX cannot be approximated with a ratio greater than $n^{\epsilon-1}$ in polynomial time for bounded $d \geq 1$ and $s \geq 2$. For $d \geq 2$, whether there is a $O(c^n)$-time exact algorithm, $c < 2$, for MAX d-BDS is still open in the literatures. For any positive integers $b \leq a$, the current best b/a-approximation algorithm for MAX d-BDS running in time $O^*(2^{bn/a})$. In this paper, we design $1/p$-approximation algorithms to solve the MAX d-BDS running in time faster than $O^*(2^{n/p})$ for integers $d \geq 2$ and $p \geq d+1$.

We end this section with some notation. A modified O-notation, O^*, is used here to bound the running time of exponential time algorithms asymptotically. For functions f and g, $f(n) = O^*(g(n))$ if $f(n) = O(g(n) \cdot poly(n))$ where $poly(n)$ is a polynomial. Given a graph $G = (V, E)$, we use $V(G)$ to denote the vertex set of G and $|V(G)| = n$. Let $N(v)$ be the set of vertices adjacent to v and $N[v] = N(v) \cup \{v\}$. For a vertex subset X, $N[X] = \bigcup_{v \in X} N[v]$ and $N(X) = N[X] \setminus X$. Let $deg(v) = |N(v)|$ denote the degree of v. For a vertex v, we use $G - v$ to denote $G[V \setminus \{v\}]$. For a vertex subset X, we use $G - X$ to denote $G[V \setminus X]$. Let $\Delta(G)$ be the maximum degree of G.

2 Inapproximability results

Chang and Hung [8] proved that if $P \neq NP$, for all $\epsilon > 0$ the MAX 1-BDS problem cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time. In this section, for bounded $d \geq 1$ and bounded $s \geq 2$, we show the inapproximability of the MAX d-BDS problem and the MAX s-PLEX problem.

Theorem 1. *If $P \neq NP$, for bounded $d \geq 1$ and all $\epsilon > 0$ the* MAX d-BDS *problem cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time.*

Proof. Suppose that the MAX d-BDS problem can be approximated to a ratio r in polynomial time, $r > n^{\epsilon-1}$ for some $\epsilon > 0$. Let S be the solution found by the approximation algorithm and S^* be an optimal solution. Then

$$\frac{|S|}{|S^*|} \geq r.$$

Let I^* be a maximum independent set in the input graph. Since I^* is also a bounded-degree-d set, $|I^*| \leq |S^*|$. There exists an independent set I of size at least $\frac{|S|}{d+1}$ in $G[S]$ by selecting $v \in S$ to be a vertex in the independent set I from S and removing all neighbors of v from $G[S]$. It is easy to see that

$$|I| \geq \frac{|S|}{d+1} \geq \frac{r \cdot |S^*|}{d+1} \geq \frac{r \cdot |I^*|}{d+1}.$$

This implies

$$\frac{|I|}{|I^*|} \geq \frac{r}{d+1} > \frac{n^{\epsilon-1}}{d+1} = n^{\epsilon'-1}$$

where $\epsilon' = \epsilon - \frac{\lg(d+1)}{\lg n} > 0$ for $n > (d+1)^{\frac{1}{\epsilon}}$.

Thus, the MAXIMUM INDEPENDENT SET problem can be approximated to a ratio greater than $n^{\epsilon'-1}$ for some $\epsilon' > 0$. This contradicts to the fact that the MAXIMUM INDEPENDENT SET problem cannot be approximated to a ratio greater than $n^{\epsilon'-1}$ in polynomial time for any $\epsilon' > 0$ if $P \neq NP$ [25]. Therefore, if $P \neq NP$, for any $\epsilon > 0$ the MAX d-BDS problem cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time. □

Corollary 1. *If $P \neq NP$, for bounded $s \geq 2$ and all $\epsilon > 0$ the* MAX s-PLEX *problem cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time.*

Proof. Suppose the MAX s-PLEX problem can be approximated to a ratio r in polynomial time where $r > n^{\epsilon-1}$ for some $\epsilon > 0$. Given a graph G, let S be the solution found by the approximation algorithm in G and S^* be an optimal solution in G. Then

$$\frac{|S|}{|S^*|} \geq r.$$

Notice that S^* is a bounded-degree-$(s-1)$ set of maximum size in $\bar{G}$ and S is a bounded-degree-$(s-1)$ set in $\bar{G}$. This implies that the MAX $(s-1)$-BDS problem can be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time, a contradiction to the fact that for bounded $d \geq 1$, the MAX d-BDS cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time if $P \neq NP$. This completes the proof. □

3 Moderately exponential time approximation algorithms

Bourgeois *et al.* [7] showed that if there exists an exact algorithm running in time $O^*(\gamma^n)$ for a maximization problem satisfying some *hereditary* property, then there exists a b/a-approximation algorithm that runs in time $O^*(\gamma^{\frac{b}{a}\cdot n})$ for the same problem for any two positive integers $b \leq a$. The problem MAX d-BDS is a maximization problem satisfying the hereditary property, *i.e.,* a subset of a bounded-degree-d set is also a bounded-degree-d set, we have the following theorem.

Theorem 2 ([7]). *If the* MAX d-BDS *problem can be solved in time $O^*(\gamma^n)$ time, then there exists an b/a-approximation algorithm that runs in time $O^*(\gamma^{\frac{b}{a}\cdot n})$ for the* MAX d-BDS *problem for any two positive integers $b \leq a$.*

Remark 1. *Chang* et al. *[9] gave an exact algorithm for* MAX 1-BDS *running in time $O^*(1.4658^n)$. Thus,* MAX 1-BDS *can be approximable to the ratio 1/2, 1/3, 1/4, and 1/5 in time $O^*(1.2108^n)$, $O^*(1.1360^n)$, $O^*(1.1004^n)$, and $O^*(1.0795^n)$, respectively. However, whether there exists $O^*(\gamma^n)$-time, $\gamma < 2$, exact algorithm for* MAX d-BDS*, $d \geq 2$, is still open. Since one can solve* MAX d-BDS *in time $O^*(2^n)$, for $d \geq 2$, there exists a b/a-approximation algorithm for the* MAX d-BDS *problem running in time $O^*(2^{bn/a})$.*

In this section, for $d \geq 2$ and $p \geq d+1$, we give $1/p$-approximation algorithms for the MAX d-BDS problem running in time faster than $O^*(2^{n/p})$ by applying three strategies.

3.1 Find a defective coloring

A graph $G = (V, E)$ is called (p, d)*-colorable* if it admits a vertex coloring with p colors such that each color class in G induces a bounded-degree-d set. Here d is called the *defect* of the coloring. Defective coloring was introduced in [1, 15, 10]. The *defective chromatic number* χ_d of a graph $G = (V, E)$ is to find the minimum number p such that G is (p, d)-colorable. It is NP-complete to determine whether a graph has a (p, d)-coloring for $p \geq 3$ and $d \geq 0$ [11]. Moreover, for $p = 2$ and $d \geq 1$, to determine whether a graph is (p, d)-colorable is also NP-complete, even for planar graphs [11]. It was shown that by reduction from ordinary vertex coloring there exists $\epsilon > 0$ such that χ_d cannot be approximated with ratio n^ϵ unless $P = NP$ [18, 11].

In this section, we show that if a graph $G = (V, E)$ can be (p, d)-colored, then there exists a bounded-degree-d set in G of size at least $\lceil n/p \rceil$.

Lemma 1. *If a graph $G = (V, E)$ can be (p, d)-colored, then there exists a bounded-degree-d set in G of size at least $\lceil n/p \rceil$.*

Proof. Since $G = (V, E)$ is (p, d)-colorable, V can be partitioned into $V_1, V_2, \ldots, V_p$ satisfying that vertices in V_i have the same color. Let $\beta = \max_{i=1}^{p} |V_i|$. It is easy to see that $\beta \geq \lceil n/p \rceil$ where n is the number of vertices in G. Notice that each V_i is a bounded-degree-d set. This shows that if G admit a (p, d)-coloring, then it has bounded-degree-d set of size at least $\lceil n/p \rceil$. □

The following result of Lovász [17] gives an upper bound of the defective chromatic number of a graph with maximum degree Δ.

Lemma 2 ([17],[11]). *For any p, any graph $G = (V, E)$ of maximum degree Δ can be $(p, \lfloor \Delta/p \rfloor)$-colored in time $O(\Delta \cdot |E|)$.*

Theorem 3. *For any p, any graph $G = (V, E)$ satisfying $\Delta(G) \leq p \cdot (d+1) - 1$ has a bounded-degree-d set of size at least $\lceil n/p \rceil$ that can be found in time $O(\Delta(G) \cdot |E|)$.*

Proof. Since $\Delta(G) \leq p \cdot (d+1) - 1$, $\lfloor \Delta(G)/p \rfloor \leq d$. According to Lemma 2, G admits a (p, d)-coloring that can be found in time $O(\Delta(G) \cdot |E|)$. By Lemma 1, the color class that has a maximum number of vertices form a bounded-degree-d set of size at least $\lceil n/p \rceil$. This completes the proof. □

3.2 Prune the search tree

In this section, for $p \geq d+1$, we give a moderately exponential time $1/p$-approximation algorithm for MAX d-BDS. The algorithm consists of a branching rule and a boundary condition. The branching rule is used to recursively solve the smaller instances of the problem with smaller instances. Search trees are often used to illustrate the execution of a branching algorithm. The root of a search tree represents the input of the problem, every child of the root represents a smaller instance reached by applying a branching rule associated with the instance of the root. One can recursively assign a child to a node in the search tree when applying a branching rule. The running time of a branching algorithm is usually measured

Problems	branching vector	Approx. ratio $1/p$	Running time
MAX 2-BDS	(1, 10)	1/3	$O^*(1.1975^n)$
MAX 3-BDS	(1, 17)	1/4	$O^*(1.1340^n)$
MAX 4-BDS	(1, 26)	1/5	$O^*(1.0976^n)$

Table 1: The running time of **Algorithm A** for MAX d-BDS, $d = 2, 3, 4$ and $p = d + 1$

by the maximum number of leaves in its corresponding search tree.

When a branching rule b is applied, the current instance is branched into $q \geq 2$ instances of size at most $n - t_1, n - t_2, \ldots, n - t_q$ where n is the number of vertices in the current instance. We call $\mathbf{b} = (t_1, t_2, \ldots, t_q)$ the *branching vector* of branching rule b. This can be formulated in a linear recurrence

$$T(n) \leq T(n - t_1) + T(n - t_2) + \cdots + T(n - t_q)$$

where $T(n)$ is the number of leaves in the search tree. The running time of the branching algorithm using only branching rule b is $O(poly(n) \cdot T(n)) = O^*(c^n)$, where c is the unique positive real root of $x^n - x^{n-t_1} - x^{n-t_2} - \cdots - x^{n-t_q} = 0$ [13]. The number c is called the *branching number* of the branching vector $(t_1, t_2, \ldots, t_q)$.

Algorithm A

- **Branching Rule.**
  If $\Delta(G) \geq p \cdot (d + 1)$, pick a vertex v of maximum degree in G and branch the following two cases.

  1. Discard v, *i.e.,* v is not in S. The algorithm recursively solves the problem on the instance $G - v$.
  2. Select v, *i.e.,* $S := S \cup \{v\}$, discard all vertices in $N(v)$, and arbitrarily discard $p - d - 1$ vertices in $V - N[v]$. The algorithm recursively solves the problem on the instance $G - (N[v] \cup X)$ where X is the set of $p - d - 1$ vertices being discarded in $V - N[v]$.

- **Boundary Condition.**
  If $\Delta(G) \leq p \cdot (d + 1) - 1$, by Theorem 3, a bounded-degree-d set of size at least $\lceil n/p \rceil$ can be found in polynomial time.

Theorem 4. *For any integer $p \geq d + 1$, MAX d-BDS can be approximated to a ratio $1/p$ in time $O^*(\alpha^n)$ where α is the branching number of the branching vector $(1, dp + 2p - d)$.*

Proof. Let S be the bounded-degree-d set returned by **Algorithm A**. Let $G' = (V', E')$ be the induced subgraph satisfying the **Boundary Condition**. Let $S = S' \uplus S''$ where S' is the part of solution found in G'. If $\Delta(G) \leq p \cdot (d + 1) - 1$, by Theorem 3, $|S'| \geq \lceil |V'|/p \rceil$ can be found in polynomial time.

If $\Delta(G) \geq p \cdot (d+1)$, the algorithm branches on a vertex v of maximum degree that either v is in S'' or v is not in S''. We may assume that there exists an optimal solution S^* such that $S'' \subseteq S^*$. For any $v \in S''$, at most d of its neighbors are in S^*. Since **Algorithm A** discards all vertices in $N(v)$ and $p-d-1$ vertices in $V - N[v]$. Suppose that all those discarded $p - d - 1$ vertices in $V - N[v]$ are in S^*. During the branching stage, once **Algorithm A** select a vertex $v \in S^*$, it would discard at most $p - 1$ vertices in S^*. Thus

$$\begin{aligned} \frac{|S|}{|S^*|} &= \frac{|S'| + |S''|}{|S^*|} \\ &\geq \frac{\frac{1}{p} \cdot |S^* \cap V'| + \frac{1}{p} \cdot |S^* - V'|}{|S^*|} \\ &= \frac{1}{p} \end{aligned}$$

For graphs satisfying the **Boundary Condition**, **Algorithm A** takes polynomial time to solve the remaining problem. We analyze the running time of the branching part of **Algorithm A**. Let $T(n)$ denote the number of leaves in the corresponding search tree. The algorithm either discards a vertex v of degree at least $p(d+1)$ or selects v and discards all vertices in $N(v)$ and $(p - d - 1)$ vertices in $V - N[v]$. Thus,

$$T(n) = T(n - 1) + T(n - pd - 2p + d).$$

We see that the running time of **Algorithm A** is $O^*(\alpha^n)$ where α is the branching number of the branching vector $(1, pd + 2p - d)$. This completes the proof. $\square$

Corollary 2. *For $p = d + 1 \geq 3$, there exists a $1/p$-approximation algorithm running in time faster than $O^*(2^{n/p})$.*

Proof. By Theorem 4, **Algorithm A** is a $1/p$-approximation algorithm with running time $O^*(\alpha^n)$ where α is the branching number of the branching vector $(1, dp+2p-d)$. For $p = d+1$, α is the branching number of the branching vector $(1, p^2+1)$. Note that α is the unique positive root of $x^{p^2+1} - x^{p^2} - 1 = 0$. Since for $p \geq 3$,

$$(2^{1/p})^{(p^2+1)} - (2^{1/p})^{p^2} - 1 > 0,$$

we obtain that $\alpha < 2^{1/p}$. This shows that for $p = d+1 \geq 3$, there exists a $1/p$-approximation algorithm running in time faster than $O^*(2^{n/p})$. $\square$

In Table 1, we list the running time of **Algorithm A** for MAX d-BDS, $d = 2, 3, 4$ and $p = d+1$.

3.3 Divide and approximate

In this section, we combine a strategy called *divide and approximate* given in [7] and **Algorithm A** to design a $1/p$-approximation algorithm for MAX d-BDS running in time faster than $O^*(2^{n/p})$ for $p > d+1$.

Algorithm B

- Partition V into $(V_1, V_2, \ldots, V_a)$ where $|V_i| \leq \lceil n/a \rceil$.
- Construct $G_1, G_2, \ldots, G_a$ where $G_i = G[\bigcup_{j=i}^{i+b-1} V_{(j \mod a)}]$.
- Run **Algorithm A** on G_i, $i = 1, \ldots, a$.
- Output the best solution among all solutions found by **Algorithm A** on G_i, $i = 1, \ldots, a$.

In the following lemma, we show that **Algorithm B** is a b/qa-approximation algorithm for MAX d-BDS.

Lemma 3. *For positive integers a, b, q satisfying $a > b$ and $q \geq d+1 \geq 3$, there exists a b/qa-approximation algorithm for* MAX d-BDS *running in time $O^*(\alpha^{bn/a})$ where α is the branching number of the branching vector $(1, dq+2q-d)$.*

Proof. Let S be the bounded-degree-d set found by **Algorithm B**, S_i^* be a maximum bounded-degree-d set in G_i, $i = 1, \ldots, a$, and $|S_a^*| = \max_{i=1}^{a} |S_i^*|$. Let S_i, $i = 1, 2, \cdots, a$ be the bounded-degree-d set found by **Algorithm A** in G_i. Suppose that S^* is a maximum bounded-degree-d set in G. We see that

$$|S^* \cap V(G_i)| \leq |S_i^*|$$

and

$$\sum_{i=1}^{a} |S^* \cap V(G_i)| \leq a \cdot \max_{1 \leq i \leq a} |S_i^*| = a \cdot |S_a^*|.$$

Since $V(G_i) = \bigcup_{j=i}^{i+b-1} V_{(j \mod a)}$, we have

$$\sum_{i=1}^{a} |S^* \cap V(G_i)| = b \cdot |S^*| \leq a \cdot |S_a^*|$$

and

$$|S^*| \leq \frac{a}{b} \cdot |S_a^*|.$$

Since S_a is the bounded-degree-d found by **Algorithm A**, according to Theorem 4, $|S_a^*| \leq q \cdot |S_a|$. Thus

$$|S^*| \leq \frac{a}{b} \cdot |S_a^*| \leq \frac{a}{b} \cdot q \cdot |S_a| \leq \frac{qa}{b} \cdot |S|$$

and

$$\frac{|S|}{|S^*|} \geq \frac{b}{qa}.$$

This shows that the approximation ratio is $\frac{b}{qa}$.

Note that **Algorithm B** calls **Algorithm A** a times. **Algorithm A** takes $O^*(\alpha^{|V(G_i)|})$-time to find a bounded-degree-d set in G_i where α is the branching number of the branching vector $(1, dq+2q-d)$. The running time of **Algorithm B** is

$$O^*(\sum_{i=1}^{a} \alpha^{|V(G_i)|}) = O^*(\alpha^{bn/a}).$$

This completes the proof. $\square$

Theorem 5. *For an integer $p > d+1 \geq 3$, there exists a $1/p$-approximation algorithm for* MAX d-BDS *running in time faster than $O^*(2^{n/p})$.*

Proof. By Corollary 2, **Algorithm A** is a $1/(d+1)$-approximation running in time $O^*(\alpha^n)$ faster than $O^*(2^{n/(d+1)})$, *i.e.*, $\alpha < 2^{1/(d+1)}$. Let $a = p$, $b = d+1$, and $q = d+1$. According to Lemma 3, there exists a b/pa-approximation algorithm for MAX d-BDS running in time

$$O^*(\alpha^{bn/a}) = O^*(\alpha^{\frac{d+1}{p} \cdot n}).$$

Since $\alpha < 2^{1/(d+1)}$, we see that $\alpha^{\frac{d+1}{p}} < 2^{\frac{1}{p}}$. This shows that for $p > d+1 \geq 3$, there exists a $1/p$-approximation algorithm for MAX d-BDS running in time faster than $O^*(2^{n/p})$. $\square$

In Table 2, we list the running time of **Algorithm B** for MAX d-BDS, $d = 2, 3, 4$. Notice that it applies a $1/(d+1)$-approximation algorithm (**Algorithm A**) as a subroutine.

Problems	q	b/a	Approx. ratio $\frac{b}{qa} = \frac{1}{p}$	Running time
Max 2-bds	3	3/4	1/4	$O^*(1.1448^n)$
		3/5	1/5	$O^*(1.1143^n)$
		1/2	1/6	$O^*(1.0944^n)$
Max 3-bds	4	4/5	1/5	$O^*(1.1059^n)$
		2/3	1/6	$O^*(1.0875^n)$
		4/7	1/7	$O^*(1.0746^n)$
Max 4-bds	5	5/6	1/6	$O^*(1.0807^n)$
		5/7	1/7	$O^*(1.0688^n)$
		5/8	1/8	$O^*(1.0600^n)$

Table 2: The running time of **Algorithm B** for Max d-bds, $d = 2, 3, 4$

4 Concluding remarks

In this paper, for $p \geq d+1 \geq 3$, we design moderately exponential time $1/p$-approximation algorithms for the Max d-bds problem running in time faster than $O^*(2^{n/p})$. For the future works, it is interesting to see whether Max d-bds, $d \geq 2$, can be approximable to a ratio $1/p$, $2 \leq p \leq d$ in time faster than $O^*(2^{n/p})$. Moreover, it is also interesting to design exact algorithms to decide whether a graph has a (p, d)-coloring running in time $O^*(c^n)$, $c < 2$.

References

[1] J. Andrew and M. Jacobson, On a generalization of chromatic number, *Congressus Numer.* **47** (1985), pp. 33–48.

[2] B. Balasundaram, Cohesive subgroup model for graph-based text mining, *Proceedings of the 2008 IEEE Conference on Automation Science and Engineering*, pp. 989–994, 2008.

[3] B. Balasundaram, S. Butenko, and I. V. Hicks, Clique relaxations in social network analysis: The maximum k-plex problem, *Operations Research* **59** (2011), pp. 133–142.

[4] B. Balasundaram, S. Chandramouli, and S. Trukhanov, Approximation algorithms for finding and partitioning unit-disk graphs into co-k-plexes, *Optimization Letters* **4** (2010), pp. 311–320.

[5] N. Betzler, R. Bredereck, R. Niedermeier, and J. Uhlmann, On bounded-degree vertex deletion parameterized by treewidth, *Discrete Applied Mathematics* **160** (2012), pp. 53–60.

[6] R. van Bevern, H. Moser, and R. Niedermeier, Approximation and tidying–a problem kernel for s-plex cluster vertex deletion, *Algorithmica* **62** (2012), pp. 930–950.

[7] N. Bourgeois, B. Escoffier, and V. Th. Paschos, Approximation of max independent set, min vertex cover and related problems by moderately exponential algorithms, *Discrete Applied Mathematics* **159** (2011), pp. 1954–1970.

[8] M.-S. Chang and L.-J. Hung, Moderately exponential time approximation algorithms for the maximum bounded-degree-1 set problem, in *Proceedings of the 30th Workshop on Combinatorial Mathematics and Computation Theory*, pp. 23–30, 2013.

[9] M.-S. Chang, L.-H. Chen, L.-J. Hung, Y.-Z. Liu, and P. Rossmanith, S. Sikdar, An $O^*(1.4658^n)$-time exact algorithm for the maximum bounded-degree-1 set problem, in *Proceedings of the 31st Workshop on Combinatorial Mathematics and Computation Theory*, pp. 9–18, 2014.

[10] L. Cowen, R. Cowen, and D. Woodall, Defective colorings of graphs in surfaces: partitions into subgraphs of bounded valence, *Journal of Graph Theory* **10** (1986), pp. 187–195.

[11] L. J. Cowen, W. Goddard, and C. E Jesurum, Defective coloring revisited, *Journal of Graph Theory* **24** (1997), pp. 205–219.

[12] M. R. Fellows, J. Guo, H. Moser, R. Niedermeier, A generalization of Nemhauser and Trotter's local optimization theorem, *Journal of Computer and System Sciences* **77** (2011), pp. 1141–1158.

[13] F. V. Fomin and D. Kratsch, *Exact Exponential Algorithms*, Springer, 2010.

[14] J. Guo, C. Komusiewicz, R. Niedermeier, and J. Uhlmann, A more relaxed model for graph-based data clustering: s-plex cluster editing, *SIAM Journal on Discrete Mathematics* **24** (2010), pp. 1662–1683.

[15] F. Harary and K. Jones, Conditional colorability II: Bipartite variations, *Congressus Numer.* **50** (1985), pp. 205–218.

[16] C. Komusiewicz, F. Hüffner, H. Moser, and R. Niedermeier, Isolation concepts for efficiently enumerating dense subgraphs, *Theoretical Computer Science* **410** (2009), pp. 3640–3654.

[17] L. Lovász, On decomposition of graphs, *Studia Math. Hung.* **1** (1966), pp. 237–238.

[18] C. Lund and M. Yannakakis, On the hardness of approximating minimization problems, *Journal of the ACM* **41** (1994), pp. 960–981.

[19] B. McClosky and I.V. Hicks, Combinatorial algorithms for the maximum k-plex problem, *Journal of Combinatorial Optimization* **23** (2012), pp. 29–49.

[20] H. Moser, R. Niedermeier, and M. Sorge, Exact combinatorial algorithms and experiments for finding maximum k-plexes, *Journal of Combinatorial Optimization* **24** (2012), pp. 347–373.

[21] N. Nishmura, P. Ragde, and D. M. Thilikos, Fast fixed-parameter tractable algorithms for nontrivial generalizations of vertex cover, *Discrete Applied Mathematics* **152** (2005), pp. 229–245.

[22] S. B. Seidman and B. L. Foster, A graph-theoretic generalization of the clique concept, *The Journal of Mathematical Sociology* **6** (1978), pp. 139–154.

[23] S. Trukhannov, *Novel approaches for solving large-scale optimization problems on graphs*, PhD Thesis, A&M University, Texas, 2008.

[24] B. Wu and X. Pei, A parallel algorithm for enumerating all the maximal k-plexes, *Proceedings of PAKDD 2007*, LNAI 4819 (2007), pp. 476–483.

[25] D. Zuckerman, Linear degree extractors and the inapproximability of max clique and chromatic number, *Theory of Computing* **3** (2007), pp. 103–138.